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Title

Activity distribution among the hamstring muscles during high-speed running: a descriptive multichannel surface EMG study

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Competing Interests

The authors have no competing interests to declare that are relevant to the content of this article.

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ABSTRACT

Purpose

This study assessed activity distribution among the hamstring muscles during high-speed running. The objective was to compare within and between muscle activity, relative contribution and hip and knee joint angles at peak muscle activity during high-speed running.

Methods

Through multichannel electromyography, we measured muscle activity in male basketball players during high-speed running on a treadmill at 15 locations: five for biceps femoris long head, four for semitendinosus, six for semimembranosus. Muscle activity was calculated for each location within each hamstring muscle individually for each percent of a stride cycle.

Results

Twenty-nine non-injured basketball players were included (mean age: 17 ± 1 years; mass, 85 ± 9 kg; height, 193 ± 9 cm). Heterogeneous activity was found for all individual hamstring muscles across multiple events of the stride cycle. In the late-swing phase, muscle activity and relative contribution of the semimembranosus was significantly higher than of the semitendinosus. There was no significant difference in hip and knee joint angles at instant of peak muscle activity, assessed locally within individual hamstring muscles, as well as in general over the whole hamstring muscle.

Conclusion

Hamstring muscles were most active in the late-swing phase during high-speed running. In this phase, the semimembranosus was most active and the semitendinosus was least active. Within the biceps femoris long head, the most proximal region was significantly more active in the late-swing phase, compared to other muscle regions. For each muscle and location, peak muscle activity occurred at similar hip and knee joint angles.

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INTRODUCTION

Hamstring injuries are the most prevalent type of injuries in sports, generally occurring during high-speed running.[1-4] It has been suggested that the bi-articular hamstring muscles (biceps femoris long head, semitendinosus and semimembranosus) are most susceptible to injury during the late-swing phase of the stride cycle of high-speed running.[5-7] According to computational models, forces produced by the hamstring muscles are peaking in the late swing, simultaneously with high velocity muscle lengthening.[8-11] These muscles are accountable for the deceleration of hip flexion and knee extension,[8-11] and thus absorb high levels of eccentric mechanical load during the late swing phase.

A high contribution of biceps femoris long head during strenuous exercise compared to the semitendinosus and semimembranosus was associated with an increased risk of a first-time injury.[12] It has been suggested that hamstring injuries occur because of an inadequate distribution of individual contributions of hamstring muscles, also known as load sharing.[12,13] Load sharing refers to the distribution of muscle forces in a synergistic muscle group to jointly produce a joint moment.[14] Forces exerted by individual muscle and thus load sharing cannot be assessed non-invasively in vivo. Distribution of muscle activity has therefore previously been used as a proxy of load sharing.[15] Previous studies successfully assessed muscle activity through surface electromyography (EMG),[16] functional magnetic resonance imaging (fMRI)[12,17,18] and shear wave elastography.[19,20] A relatively high activity of the semitendinosus may protect against injury, possibly because it would unload the most frequently affected long head of the biceps femoris.[17,21,22] Additional information regarding load sharing between hamstring muscles during high-speed running may provide new information on the aetiology of hamstring injuries.

It has previously been reported that activation patterns differ both within and between the biceps femoris long head and semitendinosus during high-speed running and various exercises.[23-26] On average, peak muscle activity during high-speed running occurs in the late swing phase, yet muscle activation patterns between hamstring muscles differ substantially between athletes.[25,27] Running speed seems to have an influence on both the amplitude and timing of muscle activity: amplitude increases with increasing running speed,[25,28] and at near maximum running speed, peak activity of the biceps femoris long head compared to the semitendinosus is delayed.[28] While the amplitude of muscle activity within individual hamstring muscles can be heterogeneous,[23-26] it is unknown whether regional peaks of this amplitude within individual muscles occur at the same hip and knee joint angles. A better understanding of peak muscle activation patterns within and between hamstring muscles is required to provide more insight into the aetiology of hamstring injuries. Furthermore, (distribution of) EMG activity within the semimembranosus over the different phases of a stride cycle during high-speed running has not previously been reported. Especially for larger muscles as the hamstrings, the application of multichannel EMG would be an appropriate method.[29]

The aims of this descriptive study were; 1) to asses muscle activity within and between the three hamstring muscles (biceps femoris long head, semitendinosus and semimembranosus) over one stride cycle during high-speed running, 2) to investigate the relative contribution between muscles, and 3) to evaluate hip and knee joint angles at instants of peak muscle activity within and between the three individual hamstring muscles.

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MATERIALS & METHODS

Participants

This study is a sub-study of a larger randomized controlled trial (RCT), in which the efficacy of two hamstring injury prevention exercises in male basketball players is evaluated by diffusion tensor MRI (Dutch trial register ID: NL7248). The main study consists of three arms of 24 participants: a *Control group*, *NHE intervention* and the *Diver hamstring exercise intervention*. Both the *Control group* and the *NHE intervention* group were invited to participate in the present study over the period of accessibility of the laboratory. Participants were recruited from basketball teams and via promotion at sports and hospital facilities as well as social media platforms. Inclusion criteria were 16 years of age or older, male, playing basketball and the exclusion criterion was a hamstring injury within the preceding 12 months. All participants gave written consent before the start of data collection. Study protocol and testing procedure were approved by the medical research ethics committee Academic Medical Center Amsterdam (NL63496.018.17) and were in accordance with the Declaration of Helsinki.

Study preparation

Multichannel electromyography & electrode placement

Muscle activity distribution was assessed through multichannel EMG. Hamstrings of the left leg were studied to align with a magnetic resonance imaging procedure in the main study. The skin of the posterior thigh was shaved and cleaned with alcohol (70%) before placing electrodes. Starting posture for electrode placement were in accordance with the SENIAM guidelines.[30] The participants were lying prone with the knee slightly flexed, to contract the hamstrings in order to distinguish muscle belly from tendon.[30] The hamstrings were individually palpated to determine proximal and distal margins of each muscle belly. The superficial anatomical position of the hamstring muscles and low volumes of subcutaneous fat tissue in the participating athletes made manual palpation adequate for identifying individual hamstring muscles.[31]

Fifteen electrode pairs (22x28 mm, Blue Sensor N-00-S, Ambu Medicotest A/S, Ølstykke, Denmark) were uniformly distributed along the proximal-distal axis of each muscle belly: five over the biceps femoris long head, four over the semitendinosus and six over the semimembranosus (Figure 1). The number of electrode pairs per muscle was chosen to cover the skin surface area evenly and in a comparable manner between participants of different body height. The most proximal and distal electrodes were placed against the proximal and distal margins, the remainder was divided with an equal distance between electrode pairs per muscle. The inter-electrode distance was 22 mm and the distance between pairs depended on the individual's muscle length. The reference electrode was placed over the left anterior superior iliac spine. Signals were differentially amplified and stored on a computer through a wired connection (Porti7-16bt, TMS International BV, Oldenzaal, the Netherlands; input impedance > 10^{12} Ω, analogue-digital conversion at 2,000 samples per second, 22-bit resolution). Cables were taped and secured to the leg with a soft foam wrap to minimize movement artefacts.

Maximal voluntary isometric contraction

Three maximum voluntary isometric contractions (MVIC) were performed for EMG normalization with approximately 30 seconds rest in between. In prone position, 15° knee flexion with the ankle in neutral

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position, the examiner fixated the foot manually. The participants gradually increased knee flexion effort from rest to maximum and sustained the maximum for approximately three seconds.[32] Participants were verbally encouraged to ensure maximal effort.[33]

Motion tracking

One Cluster marker on the dorsal side of the sacrum and two cluster markers at the lateral side of the left thigh and left calf were used to assess hip and knee flexion-extension angles. One single marker was placed on the outer side of the shoe, at the height of the lateral head of the fifth metatarsal five to assess foot ground contact. Three-dimensional coordinates were collected with three Optotrak Certus cameras (Northern Digital, Waterloo ON, Canada), surrounding a treadmill (Bonte Technology B.V., Zwolle, The Netherlands), with global orientation: X; forwards-backwards, Y; left-right, Z; upwards-downwards. The motion capture system collected samples at 100 Hz. Kinematic and EMG recordings were synchronized by a pulse, sent by the motion capture system and received as extra input channel in the EMG system.

Data collection

Participants wore a safety harness during the high-speed running trials, and were instructed to run in the centre of the treadmill throughout the whole measurement. Participants performed a five-minute warm-up, consisting of jogging on the treadmill at self-selected speed and optionally self-selected stretch exercises. After warm-up, participants were verbally instructed how to run during three experimental trials while muscle activity and kinematic data were collected. Participants were instructed to catch up with a gentle linear acceleration of the treadmill (approximately 0.88 $m.s^{-2}$), up to a running speed which they subjectively could maintain for maximally 10 seconds. Participants were instructed to verbally express when they reached this running speed, after which an examiner manually terminated the acceleration of the treadmill. The treadmill was then kept at a constant speed for approximately three seconds, after which the treadmill was decelerated.

Data analysis

Kinematic data

The trial with the highest treadmill speed and in field of view of the motion capture cameras was analysed, using MATLAB R2020b (The MathWorks, Natick, MA, United States). Three consecutive strides cycles were manually identified at the highest speed during the phase at which running speed was constant, based on characteristic changes in kinematic time series of the foot marker.[5] A distinct pattern was present in the vertical displacement of the foot marker (e.g. toe-off was identified as a constant value of vertical displacement). A stride cycle was defined from toe-off, to the next toe-off of the same foot.[34] Hip joint angles were calculated from the cluster marker on the pelvis and thigh, using Euler decomposition in the order Y-X-Z (sagittal plane flexion–coronal plane flexion–transverse plane rotation).[35] Knee joint angles were calculated from the cluster markers on the thigh and calf, using Euler decomposition in the order Y–X–Z (sagittal plane flexion–coronal plane flexion–transverse plane rotation).[35] Per stride cycle, three phases were determined: early-swing; from toe-off to maximum knee flexion, late-swing; from maximum knee flexion to heel-strike, stance; from heel-strike to toe-off.

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Electromyographic data

For multichannel EMG data, both MVIC and run recordings were processed with a bi-directional second order Butterworth band-pass filter of 25-500Hz, rectification and a bi-directional second order Butterworth low-pass filter of 25 Hz. The maximum value of a single sample of EMG activity in the three MVIC attempts of each electrode location, was used for normalization of the signal of the corresponding electrode location during the run trial (EMG in percentage maximal voluntary isometric contraction, %MVIC).[36] Different cutoff frequencies for the band-pass filter are presented in Supplementary Figure 1.

Individual phases of three consecutive strides were extracted from the multichannel EMG. Data were time normalized per stride cycle (100 data samples over a complete stride, 0– 100%), with the relative duration of the early-swing, late-swing and stance phase set at a fixed percentage per phase, determined as the ratio of the group average duration of the individual phases with respect to the total stride duration. The group averaged absolute durations of the three phases were 0.22 ± 0.02, 0.20 ± 0.02 and 0.11 ± 0.01 s for the early-swing, late-swing and stance phase respectively. The stride cycle (1-100%) was accordingly divided in the following relative durations: 1-42% stride cycle for the early-swing phase, 43-80% stride cycle for the late-swing phase and 81-100% stride cycle for the stance phase. Detailed results of individual absolute durations are presented in Supplementary Table 1. Periods with evident contamination (e.g. movement artefacts) in the raw EMG data over the three strides were manually labelled as NaN (not a number) in Matlab to exclude these periods from analysis. Results were averaged over three strides. Per percentage of the stride cycle, muscle average EMG activity were calculated across all electrode locations of the corresponding hamstring muscles (in %MVIC). The relative contribution of individual muscles was assessed as the ratio of the mean normalized of the biceps femoris long head, the semitendinosus and the semimembranosus, to the summated normalized muscle activity of the three muscles (relative contribution (%con) = [individual normalized muscle activity]/[summed normalized muscle activity of all three muscles]).[17,37]

Peak muscle activity

Peak EMG activity was calculated over the stride cycle 1) within muscles: at the instant that peak activity occurred for each electrode location per muscle, 2) between muscles: at the instant that the peak activity occurred in the mean activity across the electrode locations per muscle, 3) over the total hamstring muscle: at the instant that the peak activity occurred in the mean over all 15 electrode locations. For each time-point of peak EMG activity, the associated hip and knee joint angles were extracted from the kinematic time series, in degrees.[27] Full leg extension was 0°, with positive values for flexion of the hip and knee joint.

Statistical analysis

One-dimensional statistical parametric mapping (SPM) was used to test for differences in muscle activity between electrode locations within individual muscles and between muscles.[38] The whole stride cycle (1-100%) was used as one input. Repeated measures ANOVAs for one-dimensional measures were used to test for differences between electrode locations for each muscle individually, with electrode location as a factor and normalized muscle activity (%MVIC) as dependent variable. Repeated measures ANOVAs

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were used to test for differences between muscles over the stride cycle, with muscle as a factor and mean whole muscle normalized muscle activity (%MVIC) as dependent variable. Also the muscles' individual contribution was tested with a repeated measures ANOVA, with muscle as factor and relative contribution (%con) as dependent variable. In all tests, an F-ratio was estimated for each percent of the stride cycle by the SPM Matlab tool and referenced to a critical F-ratio (F*) with an alpha of 0.05. *Post hoc* analyses were applied in case of a significant difference, using paired samples t-tests for all combinations with Bonferroni corrections.

Statistical analysis on the occurrence of EMG peak moments were performed using IBM SPSS Statistics (IBM SPSS Statistics for Windows, Version 27.0. Armonk, NY: IBM Corp). One-way ANOVA with repeated measures was used to test for differences between hip and knee joint angles between EMG peak moments (repeated within-subjects factors: Peak EMG [biceps femoris long head within, semitendinosus within, semimembranosus within, mean biceps femoris long head, mean semitendinosus, mean semimembranosus, mean hamstring muscles]) and degrees as dependent variable. A Shapiro-Wilk test was used to test for normality distribution. Alpha was set to 0.05, and in case of a significant main effect, *post hoc* tests were applied with Bonferroni corrections. Hip and knee joint angles at peak activity were described as means and standard deviations.

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RESULTS

Data collection & participant characteristics

The inclusion period for this study was 13 months, while the window of inclusion for the main study was 19 months. Within this 13-months period, 38 of the total included 48 participants of the larger RCT could participate. One of the 38 participants did not show up at the scheduled time slot. Eight measured data sets were not suitable for analysis, as pelvis and foot markers were recorded improperly. Data sets of 29 participants were used for analysis. The mean age was 17 ± 1 yr.; mass, 85 ± 9 kg; height, 193 ± 9 cm; maximal running speed, 7.6 ± 0.5 $m.s^{-1}$. Two participants suffered an anterior cruciate ligament (ACL) injury of the left leg in the past: one treated non-surgically and one with a semitendinosus tendon graft reconstruction. Four cases of evident contamination were observed in the raw EMG data sets of four individual participants: once the second most distal electrode location of the semitendinosus, once the most proximal electrode location of the semimembranosus and twice the most distal electrode location of the semimembranosus. The episodes containing these contaminations were discarded and are illustrated in Supplementary Figures 2 to 5.

Distribution of activity within muscles

Within the biceps femoris long head, there was a significant main effect of electrode location on normalized muscle activity ($df(4,112)$, $F = 4.68$, $p < 0.05$, Figure 2A-B). This effect occurred in the early-swing, late-swing and stance phases. *Post hoc* tests revealed that the most proximal electrode location had a higher normalized muscle activity compared to all other electrode locations. Both the second most distal and most distal electrode location had a higher normalized muscle activity compared to the second most proximal and middle electrode in the early-swing phase. A more detailed description of the *post hoc* tests for the biceps femoris long head is illustrated in Supplementary Figure 6.

Within the semitendinosus, there was a significant main effect of electrode location on normalized muscle activity ($df(3,84)$, $F = 5.65$, $p < 0.05$, Figure 2C-D). *Post hoc* tests revealed that the second most proximal electrode location had a lower normalized muscle activity compared to the most proximal and second most distal electrode location in the late-swing phase. A more detailed description of the *post hoc* tests for the semitendinosus is illustrated in Supplementary Figure 7.

Within the semimembranosus, there was a significant main effect of electrode location on normalized muscle activity ($df(5,140)$, $F = 4.19$, $p < 0.05$, Figure 2E-F). This effect occurred in the early-swing and late-swing phase. *Post hoc* tests revealed that the most distal electrode locations had a higher normalized muscle activity compared to all other electrode locations in the early swing-phase. The second most distal electrode location had a higher normalized muscle activity compared to the third most proximal, second most proximal and third most distal electrode location in the early-swing phase. A more detailed description of the *post hoc* tests for the semimembranosus is illustrated in Supplementary Figure 8. Individual hamstring muscle distributions per participant are illustrated in Supplementary Figure 11.

Distribution of activity between muscles

Between muscles, the overall normalized muscle activity over one stride cycle during high-speed running differed significantly in the early-swing phase, late-swing phase and over the transition between stance to early-swing phase ($df(2,56)$, $F = 7.31$, $p < 0.05$, Figure 3A-B). *Post hoc* tests revealed that the

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semitendinosus had a significantly lower normalized muscle activity than the biceps femoris long head and the semimembranosus in the early-swing phase. The semitendinosus had also a significantly lower normalized muscle activity than the semimembranosus in the late-swing phase. A more detailed description of the *post hoc* tests for the mean activity between muscles is illustrated in Supplementary Figure 9.

The relative contribution between hamstring muscles over one stride cycle during high-speed running differed significantly between muscles (df(2,56), $F = 7.28$, $p < 0.05$, Figure 4A-B). *Post hoc* tests revealed that the relative contribution of the semitendinosus was lower compared to the biceps femoris long head and the semimembranosus during the early-swing and late-swing phase. A more detailed description of the *post hoc* tests for the relative contribution between muscles illustrated in Supplementary Figure 10.

Joint angles at peak EMG activity

The mean hip and knee angles at occurrence of peak EMG activity were 65.4 ± 15.0° and 57.3 ± 22.9° respectively. There were no significant differences in hip ($F(6,28) = 1.0$, $p = 0.418$) and knee ($F(6,28) = 1.6$, $p = 0.146$) joint angles at moment of peak EMG activity within and between muscles. Detailed results of the joint angles are presented in Table 1. Descriptive results are described in Supplementary Table 2 and Supplementary Figure 12.

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DISCUSSION

In this study, hamstring muscle activity and relative contribution were assessed with multichannel EMG over a stride cycle during high-speed running in uninjured male basketball players. The main findings were that during the late-swing phase (i) when normalized muscle activity was the highest, the semimembranosus muscle activity was significantly higher than the semitendinosus, (ii) there was heterogeneous activity within the biceps femoris long head, the semitendinosus and the semimembranosus, and (iii) peak EMG activity occurred at comparable hip and knee joint angles for all three hamstring muscles.

Compared to literature

Hamstring muscle activity during high-speed running was previously assessed with multichannel EMG for the biceps femoris long head and semitendinosus.[23,25] These two studies investigated primarily the effect of running speed on the amplitude of normalized muscle activity within each muscle.[23,25] One of these studies reported within hamstring muscle activity distribution as a sub-analysis.[25] In contrast to our findings, no differences were found among three regions within the biceps femoris long head and semitendinosus, when comparing muscle activity over a stride cycle during high-speed running.[25] One of multiple possible explanations is that heterogeneous activity was possibly averaged out in the aforementioned study, as means were calculated over multiple EMG channels to form three regions per muscle. Our results indicate heterogeneous muscle activity within the hamstring muscles if assessed using a configuration with multiple regions during high-speed running. There was also a difference in what MVIC was used for normalization of the signal, which probably resulted in lower normalized EMG activity in our study. The maximum value of a single sample was used in our study probably resulted in a relatively lower normalized EMG activity (Figure 2 & 3) compared to an averaged value over a 1-s epoch in the aforementioned study.[25] For comparisons between muscles, existing literature is limited to single channel EMG studies.[28,39,40] None of these studies evaluated the semimembranosus activity.[28,39,40] In the early-swing phase, studies described a significantly higher muscle activity in the semitendinosus compared to the biceps femoris long, which is in contrast to our results.[28,39,40] A possible explanation can be the heterogeneous activity distribution within the individual hamstring muscles. We measured higher normalized muscle activity in the proximal and distal regions of the biceps femoris long head during the early-swing phase compared to the middle electrode location, which roughly corresponds to the electrode location used in the aforementioned studies.[28,39,40] In the late-swing phase, no significant differences between the muscle activity of the biceps femoris long head and semitendinosus during high-speed running were reported, which is in line with our results.[28,39,40] An interesting observation in our results was the decrease of mean normalized muscle activity of all three hamstring muscles just before heel-strike, at the end of the late-swing phase (Figure 3). Here, the semitendinosus and semimembranosus activity decreased rapidly, while the biceps femoris long head remained more active into the stance phase. This effect was statistically significant when comparing relative contributions between muscles (Figure 4), yet not when looking at normalized muscle activity (Figure 3). The higher contribution of the biceps femoris long head in the late-swing phase was possibly again caused by the heterogeneous activity of this muscle. The most proximal electrode location showed again significantly higher normalized muscle activity in the late-swing phase compared to the two distal electrode location. This proximal location corresponds to the most frequently injured location within the biceps femoris

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long head.[41,42] This is also the phase during high-speed running and location within the biceps femoris long head with the highest strain according to model studies.[43,44] It would be interesting to examine neural drive between the hamstring muscles in the late-swing phase during high-speed running and the relationship between activity levels in the proximal region of the biceps femoris long head and injury occurrence.[45,46]

Hip and knee joint angles at peak EMG activity for different electrode location did not differ significantly. All peak EMG activities, both within individual muscles as well as over the mean of each of the muscles, occurred in the late-swing phase. These results corresponds to results of a previous study in which the peak EMG activity was compared between sprinting and a variety of hamstring exercises for individual hamstring muscles.[27] The reported hip and joint angles at peak EMG activity of the biceps femoris long head, semitendinosus and semimembranosus were within one standard deviation of our results.[27] All peak EMG activities occurred at high musculotendinous length, when the hamstring muscles perform the largest amount of negative work.[6] This indicates that high muscle activity, muscle strain and negative power coincide and may explain why the late-swing phase is associated with hamstring injury occurrence.

The strength of this study is the use of multichannel EMG, compared to single electrode usage. It has previously been shown that heterogeneous muscle activity can cause an over- or underestimation when using conventional single-channel EMG recordings.[24,26] To highlight the importance, we repeated the between hamstring analyses with the electrode location per muscle which corresponded most to the SENIAM guidelines.[30] As a result, the between-muscle difference in normalized muscle activity in the late-swing phase disappeared (see Supplementary Figure 13). This observation, especially because it appears in the late-swing phase, signifies the additional value of using multiple EMG channels per muscle. Future studies should confirm this by comparing multichannel EMG with SENIAM guidelines.

Limitations

One of the limitations of this study is that inclusion was restricted to uninjured basketball players, which might limit generalizability to other and injured athletes. Furthermore, an inherent limitation of using EMG is cross-talk between muscles. With multichannel EMG, electrode locations are closer to proximal and distal muscle borders compared to conventional EMG, which may be more prone to crosstalk. Given the expertise in clinical care and based on comparisons with MRI in the larger RCT, manual palpation was considered accurate for identifying hamstring muscles. By careful electrode placement over the targeted muscle belly, minimizing inter-electrode to 22 mm and using differential amplification, cross-talk and electrode placement over tendon was minimized as much as possible, but cannot be completely ruled out. It is furthermore unknown to what extent the influence is of the tendinous inscription within the semitendinosus and electrode displacement by skin displacement because of movement.[47] Future studies should consider obtaining multiple MVIC values for normalization over a variety of hip and knee joint angles, as a single MVIC can result in a fractional under- or overestimation of normalized muscle activity.[48] The approximately 4 kg extra weight of the safety harness and EMG registration device, attached on the backside of the harness, had an unknown effect on the posture of the participants during high-speed running. We did not exclude the participants with a history of a hamstring injury, older than 12 months, or participants who had a history of ACL reconstruction with semitendinosus tendon graft. They might have long-lasting effects on hamstring muscle activity and should be

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considered when assessing unilateral muscle activity.[49-52] Finally, participants were tested on a treadmill and not in overground running to allow data acquisition. Treadmill-based running analysis is comparable to over ground running and had as benefit that running speed can be regulated effectively.[53]

Conclusion

This study comprehensively described the distribution of hamstring muscle activity during high-speed running, both within and between muscles. Our findings were that (i) when the hamstring muscles were most active during high-speed running, in the late-swing phase, the semimembranosus was most active, (ii) contrasting, the semitendinosus was least active in the late-swing phase during high-speed running, (iii) within the biceps femoris long head, the most proximal region was significantly more active in the late-swing phase, compared to other muscle regions, and (iv) peak muscle activity, assessed locally within individual hamstring muscles, as well as in general over the whole muscle, occurred at similar hip and knee joint angles during high-speed running.

Perspective

The present study provides novel information about the distribution of muscle activity of the hamstring muscles during high-speed running. This comprehensive study showed that when the hamstring muscles are most active, in the late-swing phase, the level of muscle activity of the semimembranosus is significantly higher compared to the semitendinosus. Also in the late-swing phase during high-speed running, the level of muscle activity in the most proximal part of the biceps femoris long head was significantly more active compared to other regions within the same muscle. This location within the biceps femoris long head corresponds to the most injurious area of the hamstring muscle. Peaks in muscle activity occurred at similar hip and knee joint angles during high-speed running. Future studies should examine if preventive interventions have an effect on the activity distribution during high-speed running.

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STATEMENTS & DECLARATIONS

Acknowledgments

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Ethics approval

This trial was prospectively registered in the Netherlands Trial Register (date of registration 2018-07-13, ID: NL7248,). Study protocol and testing procedure were approved by the medical research ethics committee of the Academic Medical Center Amsterdam (NL63496.018.17) and were in accordance with the Declaration of Helsinki.

Consent for publication

Informed consent was obtained from all individual participants included in the study.

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FIGURE CAPTION LIST

Figure 1

Figure 1. Posterior view of the left upper-leg showing the electrode placement over the hamstring muscles and the corresponding hamstring locations. On the lateral side, electrodes in *red*: the biceps femoris long head (BF) is covered by five bi-polar electrodes (locations, *p*: most proximal, *pp*: second most proximal, *mid*: middle, *dd*: second most distal, *d*: most distal). In the middle, in *green*: the semitendinosus (ST) covered by four bi-polar electrodes (locations, *p*: most proximal, *pp*: second most proximal, *dd*: second most distal, *d*: most distal). On the medial side, in *blue*: the semimembranosus (SM) covered by six bi-polar electrodes (locations, *p*: most proximal, *pp*: second most proximal, *ppp*: third most proximal, *ddd*: third most distal, *dd*: second most distal, *d*: most distal).

Figure 2

Figure 2. Within hamstring muscles: group averaged mean muscle activity per electrode location per muscle during over a stride during high-speed running. Early-swing phase: toe-off to maximal knee flexion, 1-42%; Late-swing phase: maximal knee flexion to heel-strike, 43-80%; Stance phase: heel-strike to toe-off, 81-100%. **2A.** biceps femoris long head. **2C.** semitendinosus. **2E.** semimembranosus. **2A-C-E.** The *solid lines* represent the group averaged mean muscle activity (expressed in percentage of the maximal voluntary isometric contraction, %MVIC) per electrode location per percentage of the stride cycle of 29 participants. *Semi-transparent coloured areas* represent plus/minus one standard deviation of the group averaged mean muscle activity. Abbreviations for the electrode locations: *p*: most proximal, *pp*: second most proximal, *ppp*: third most proximal, *mid*: middle, *ddd*: third most distal, *dd*: second most distal, *d*: most distal. **2B-D-F.** The *solid black line* is the F-ratio per percentage, determined by repeated measures ANOVA. The *dashed horizontal line* represents the critical F-ratio (F*).

Figure 3

Figure 3. Between hamstring muscles: group averaged muscle activity per hamstring muscle over a stride cycle during high-speed running. Early-swing phase: toe-off to maximal knee flexion, 1-42%; Late-swing phase: maximal knee flexion to heel-strike, 43-80%; Stance phase: heel-strike to toe-off, 81-100%. **3A**. The *solid lines* represent the group averaged muscle activity (expressed in percentage of the maximal voluntary isometric contraction, %MVIC) per muscle per percentage over a stride cycle during high-speed running of 29 participants. *Semi-transparent coloured areas* represent plus/minus one standard deviation of the group averaged muscle activity. **3B.** The *solid black line* is the F-ratio per percentage, determined by a repeated measures ANOVA. The *dashed horizontal line* represents the critical F-ratio (F*).

Figure 4

Figure 4. Between hamstring muscles: group averaged relative contribution per muscle over a stride cycle during high-speed running. Early-swing phase: toe-off to maximal knee flexion, 1-42%; Late-swing phase: maximal knee flexion to heel-strike, 43-80%; Stance phase: heel-strike to toe-off, 81-100%. **4A**. The *solid lines* represent the group averaged relative contribution (normalized muscle activity per muscle expressed in percentage of the summed muscle activity, %con) per percentage over a stride

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cycle during high-speed running of 29 participants. *Semi-transparent coloured areas* represent plus/minus one standard deviation of the group averaged relative contributions. **4B.** The *solid black line* is the F-ratio per percentage, determined by a repeated measures ANOVA. The *dashed horizontal line* represents the critical F-ratio (F*).

Table 1. Group average hip and knee joint angles at occurrence of peak EMG activity during high speed running.

Peak EMG	Muscle	Hip joint angle	Knee joint angle
Single channel per muscle	Biceps femoris long head	65.3 ± 18.2	65.1 ± 28.4
	Semitendinosus	69.0 ± 9.3	63.3 ± 20.1
	Semimembranosus	66.5 ± 17.3	58.8 ± 22.3
The mean across corresponding channels per muscle	Biceps femoris long head	62.9 ± 15.5	54.9 ± 26.2
	Semitendinosus	61.3 ± 17.6	52.2 ± 22.0
	Semimembranosus	65.1 ± 13.2	52.9 ± 19.0
The mean of all channels	Hamstring muscle	67.9 ± 11.7	54.0 ± 20.0

Values are means ± standard deviations, all reported in degrees, positive values represent flexion from neutral position.

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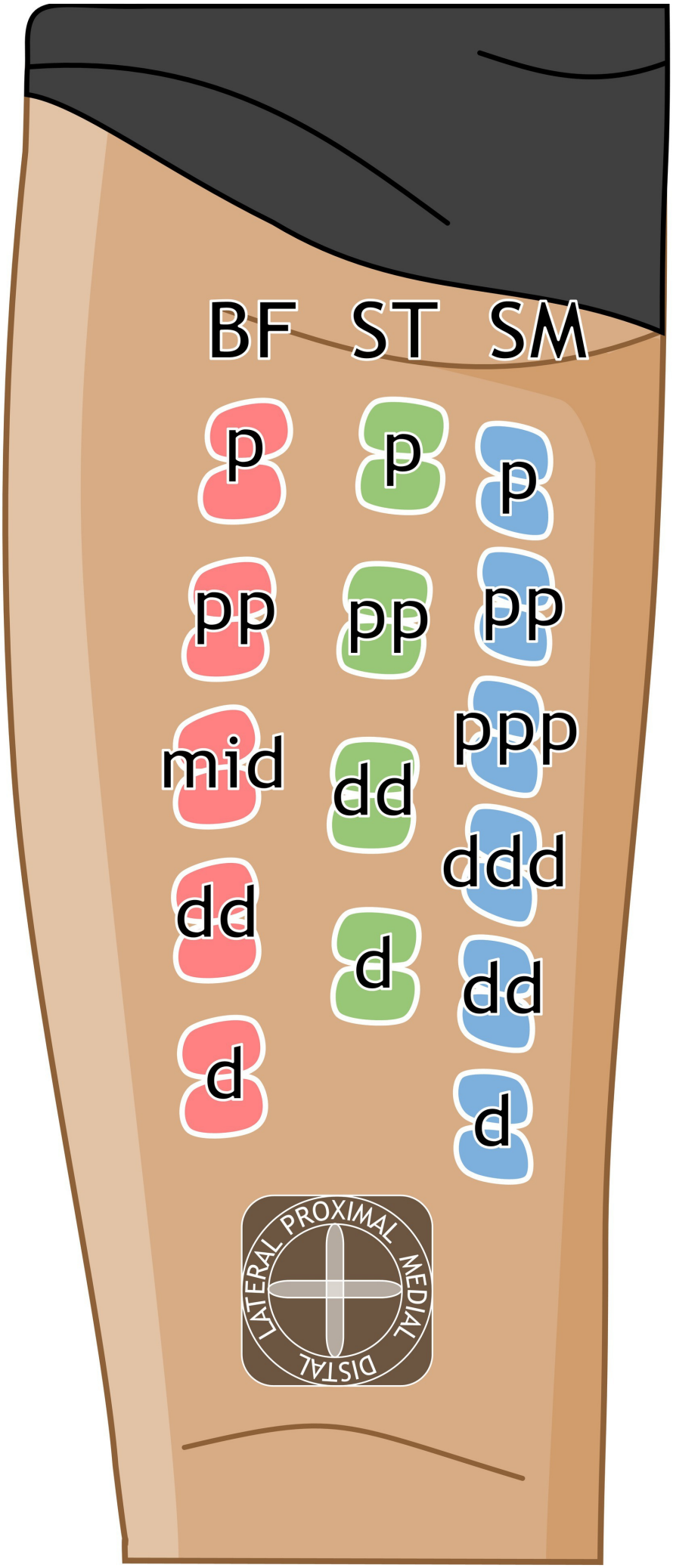

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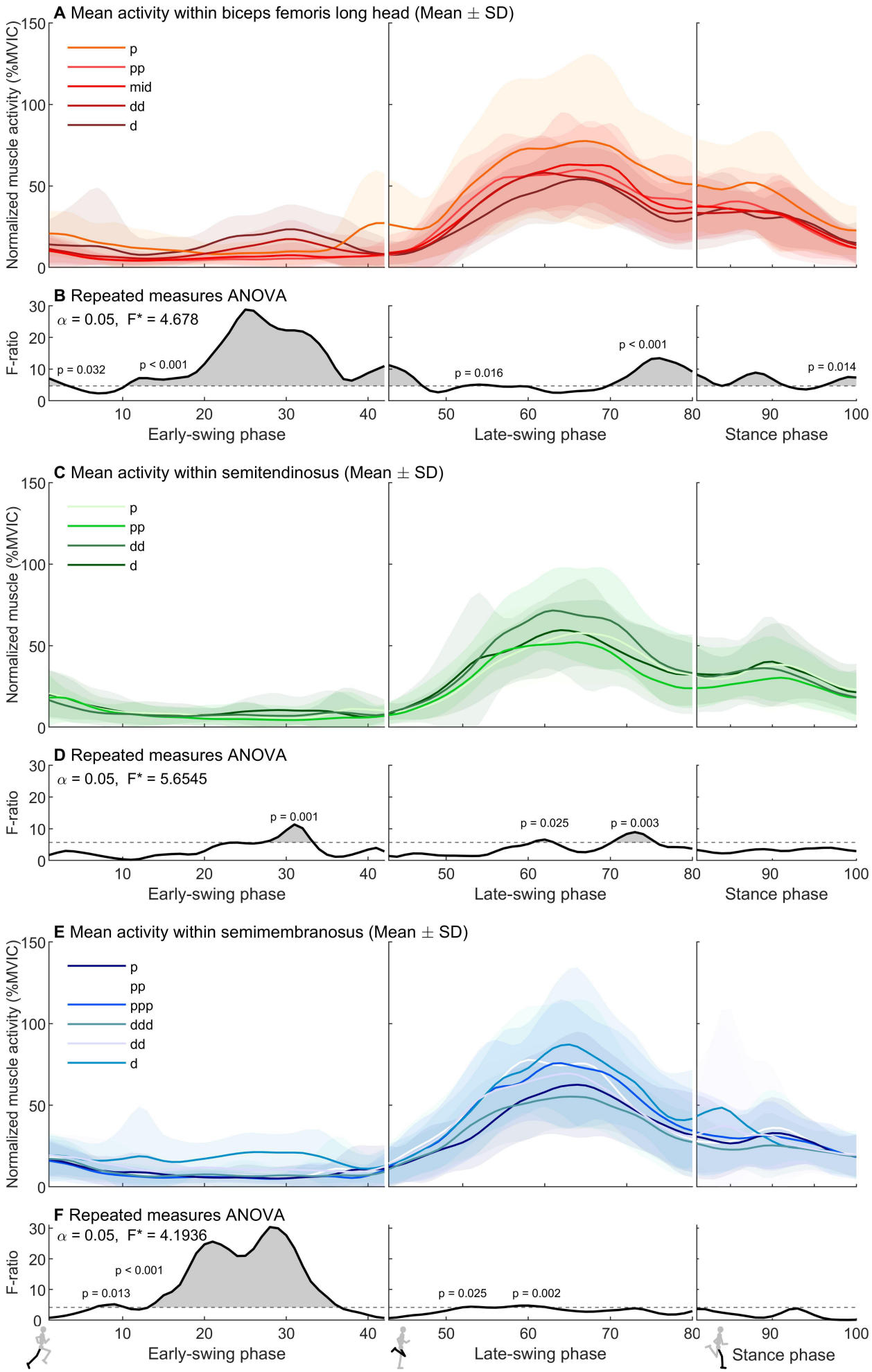

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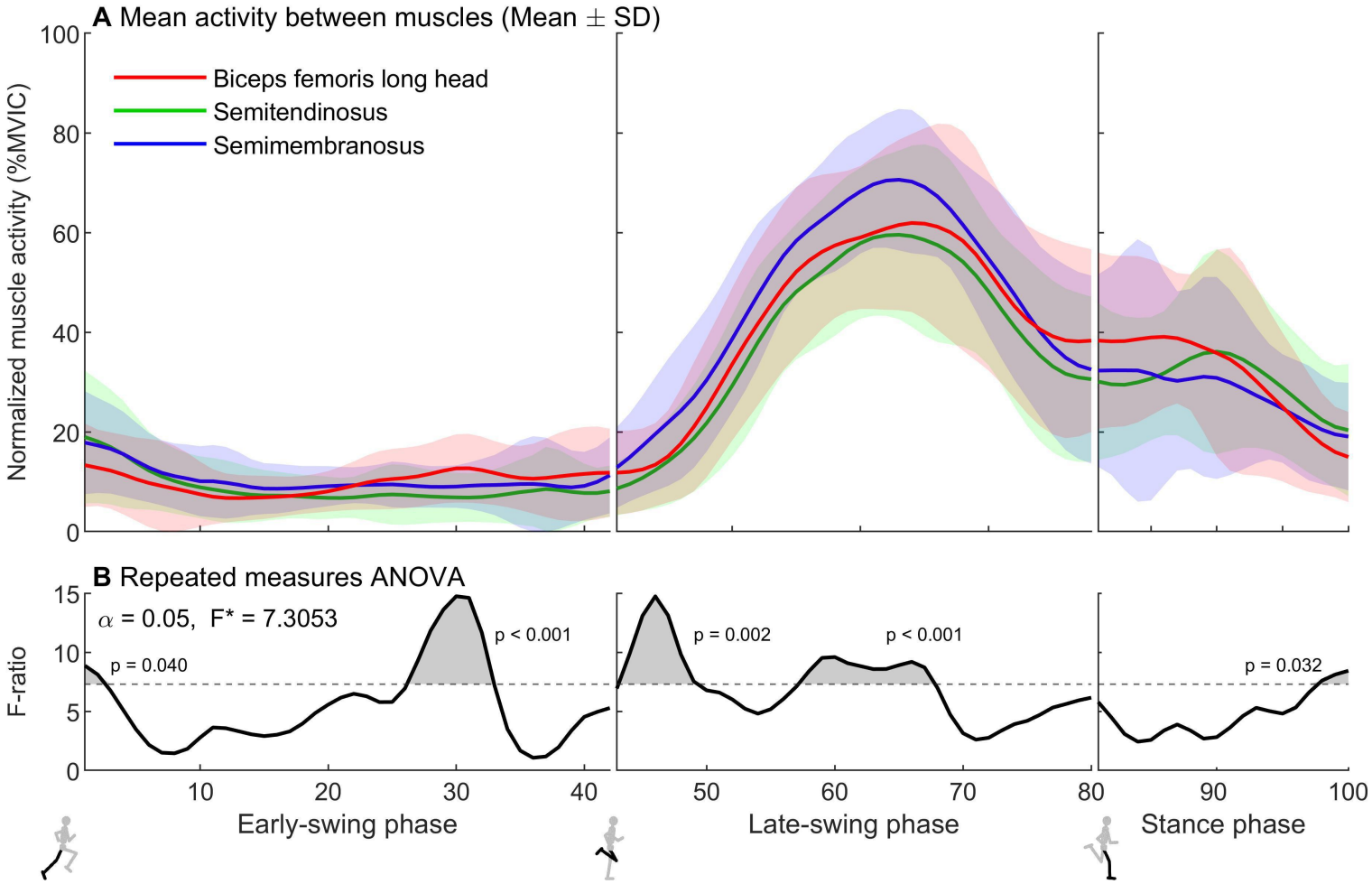

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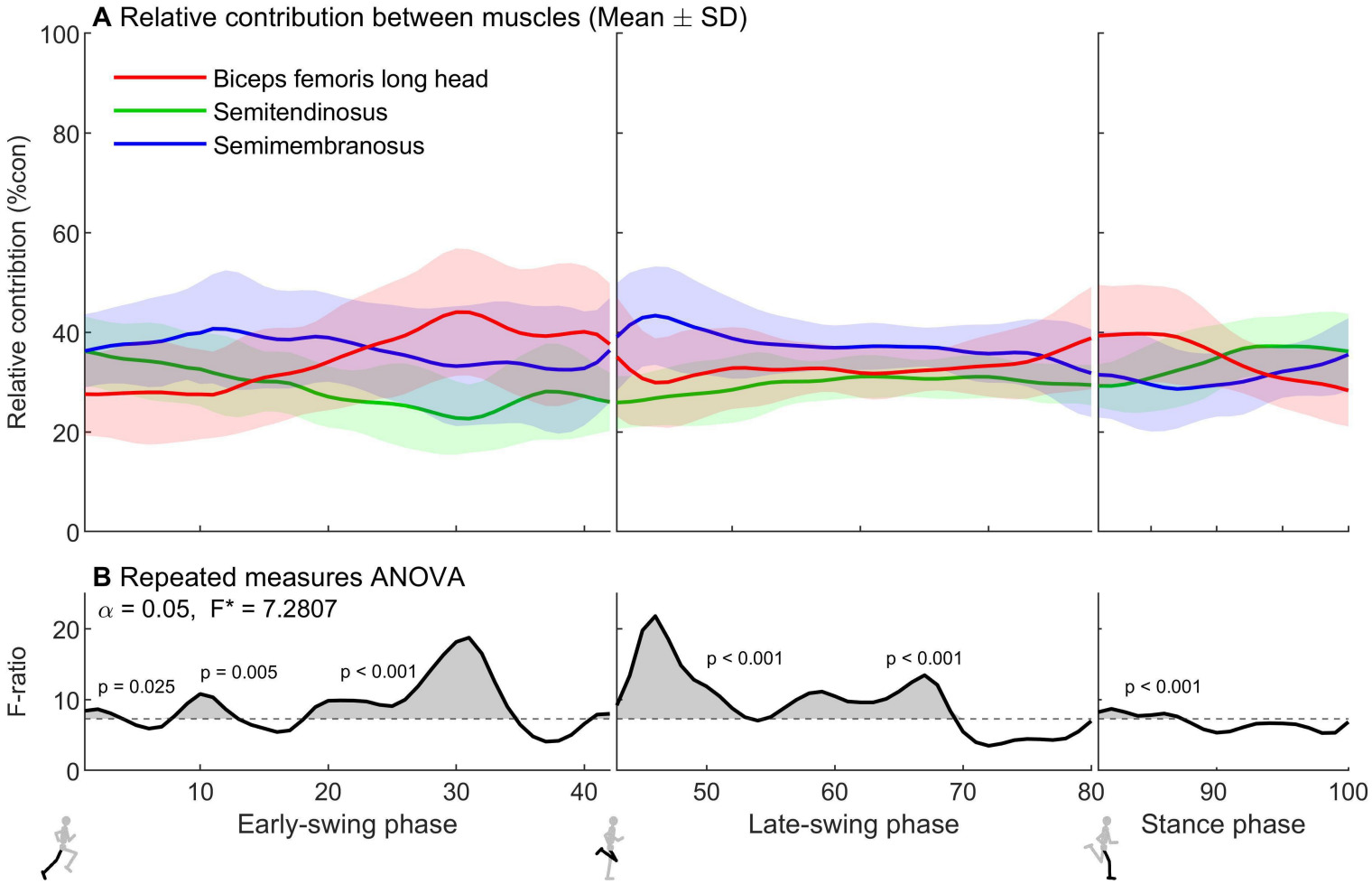

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